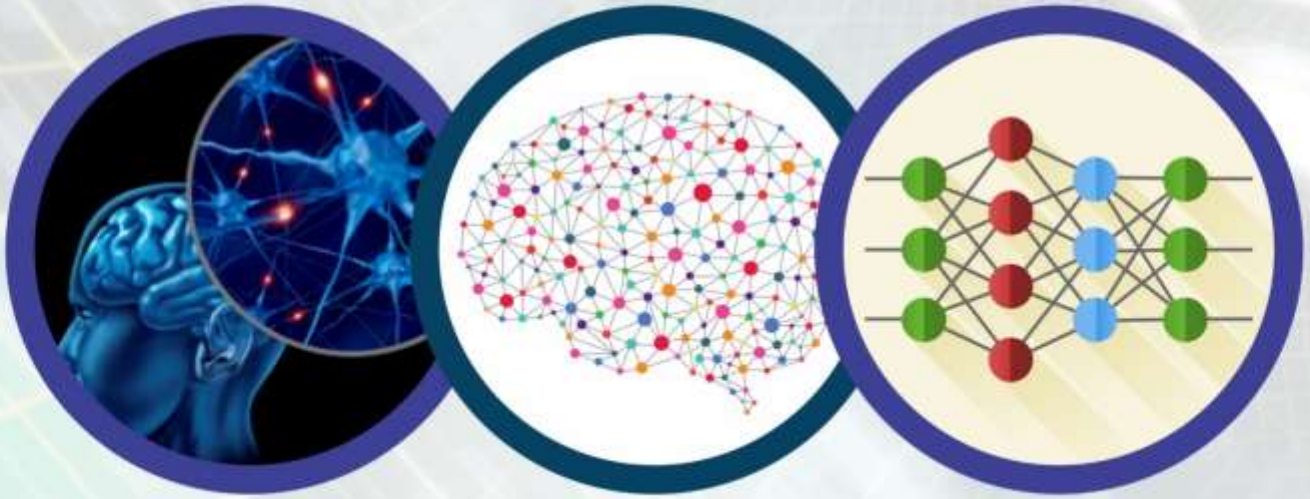
DEEP LEARNING: LEARNING BASED ON DEEP NEURAL NETWORK



Seminar Presentation by:

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OUTLINE

- Introduction
- Aim and Objectives
- Review of Deep Learning Concept
- Review of Related Works & Specific Application Areas
- Practical Application of Deep Learning (Problem Scenario)
- Trends & Research Issues
- Conclusion
- References
- Acknowledgment

INTRODUCTION

(Deep Learning: A transitive subset of Artificial Intelligence Via Machine Learning

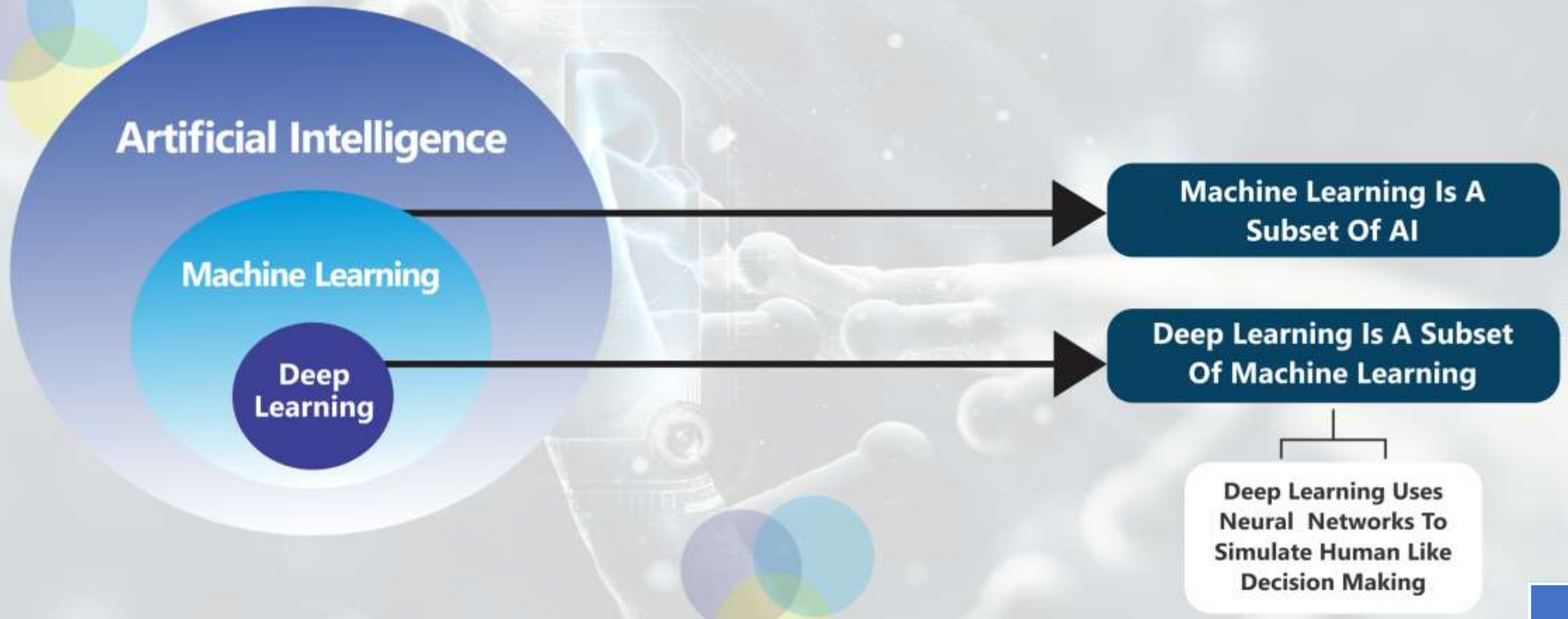


Figure 1 : Subsets of AI(Adapted from: www.edureka.co)

What is Artificial Intelligence (AI)?

- The theory and development of computer systems that are able to perform task normally requiring human intelliegence such as visual perception, speech recognition, decision-making and translation between languages

AI INCLUDES FOLLOWING AREAS OF SPECIALIZATION

- Game playing
- Expert System
- Robotics
- Neural networks
- Natural Language Processing

CHESS PLAYING GAME

NUMBER PLATE RECOGNITION

SIRI

Figure 2: Some Applications of Artificial Intelligence

AI Technologies Timeline

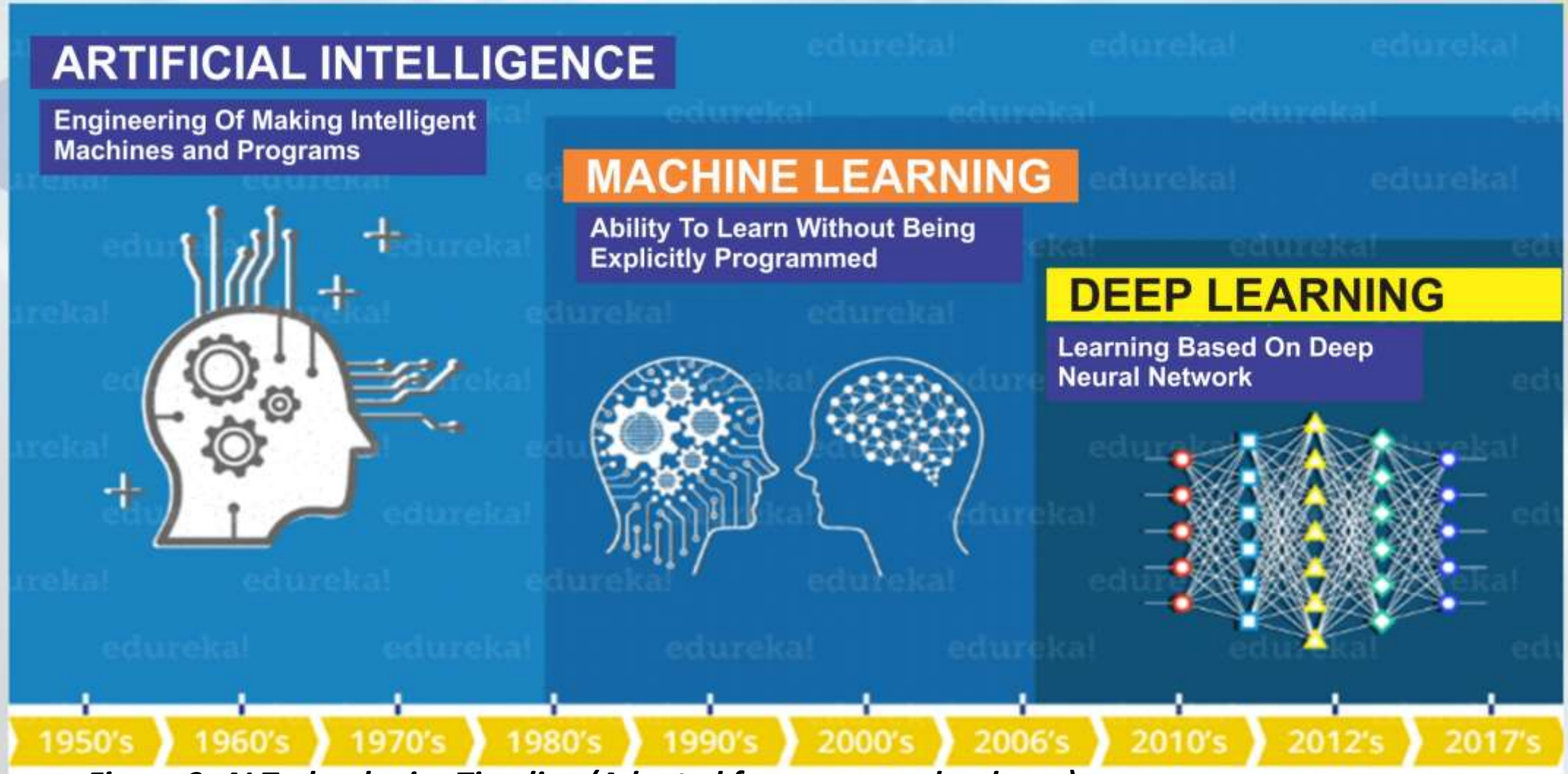


Figure 3: AI Technologies Timeline (Adapted from: www.edureka.co)

What Is Machine Learning?

- Machine learning is a type of artificial intelligence that provide computers with the ability to learn without being explicitly programmed

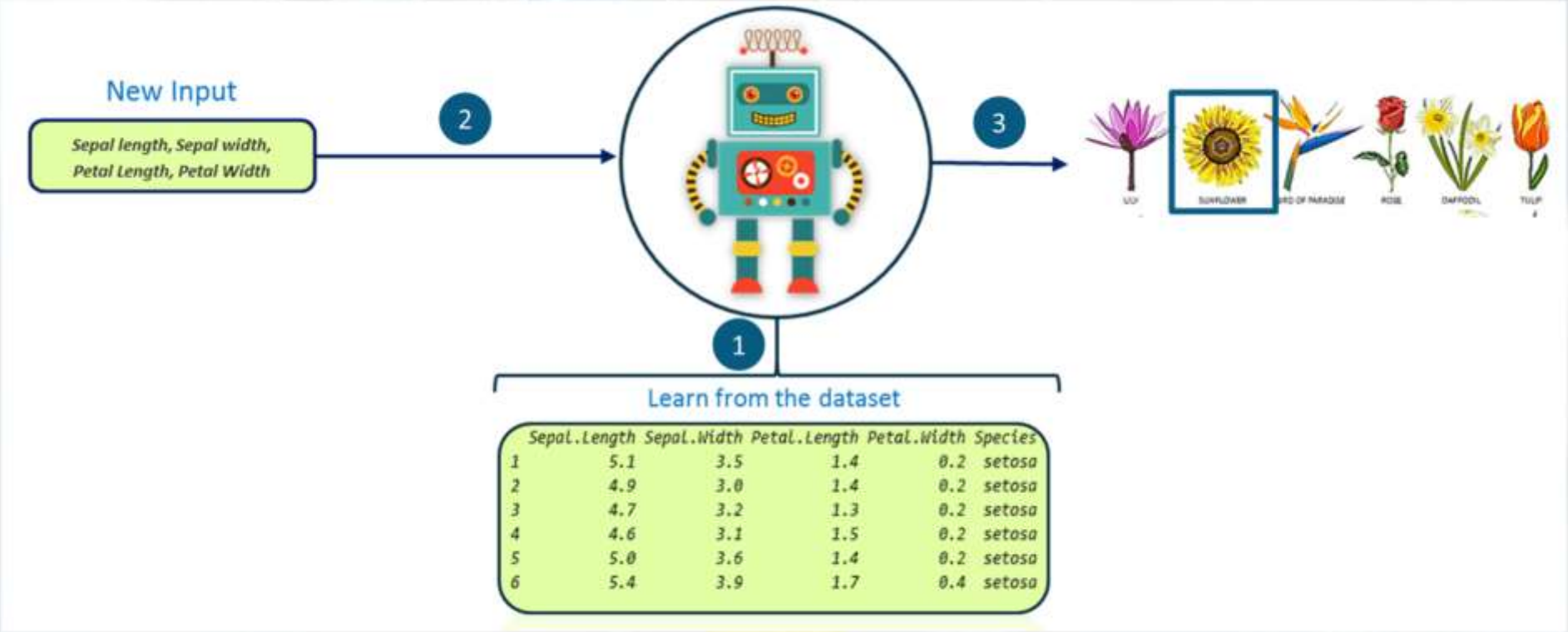


Figure 4: Process Involved in Machine Learning (Adapted from: www.edureka.co)

Limitations of Machine Learning

One Of The Big Challenges With Traditional Machine Learning Models Is A Process Called Feature Extraction. For Complex Problems Such As Object Recognition Of Handwriting Recognition, This Is A Huge Challenge.

Deep Learning To The Rescue

Deep Learning Modes Are Capable Of Focus On The Right Features By Themselves, Requiring Little Guidance From The Programmer.
These Models Can Also Partially Solve The Dimensionality Problem.

The Idea Behind Deep Learning Is To Build Learning Algorithms That Mimic Brain

Figure 5: Limitation of ML (Source: www.edureka.co)

The aim of this seminar presentation is to review the concept, architecture, algorithms, trends and research issues in deep learning.

1. To discuss the concept of deep learning by way of definition, architecture and basic algorithms.

2. To review related works and highlight specific application areas

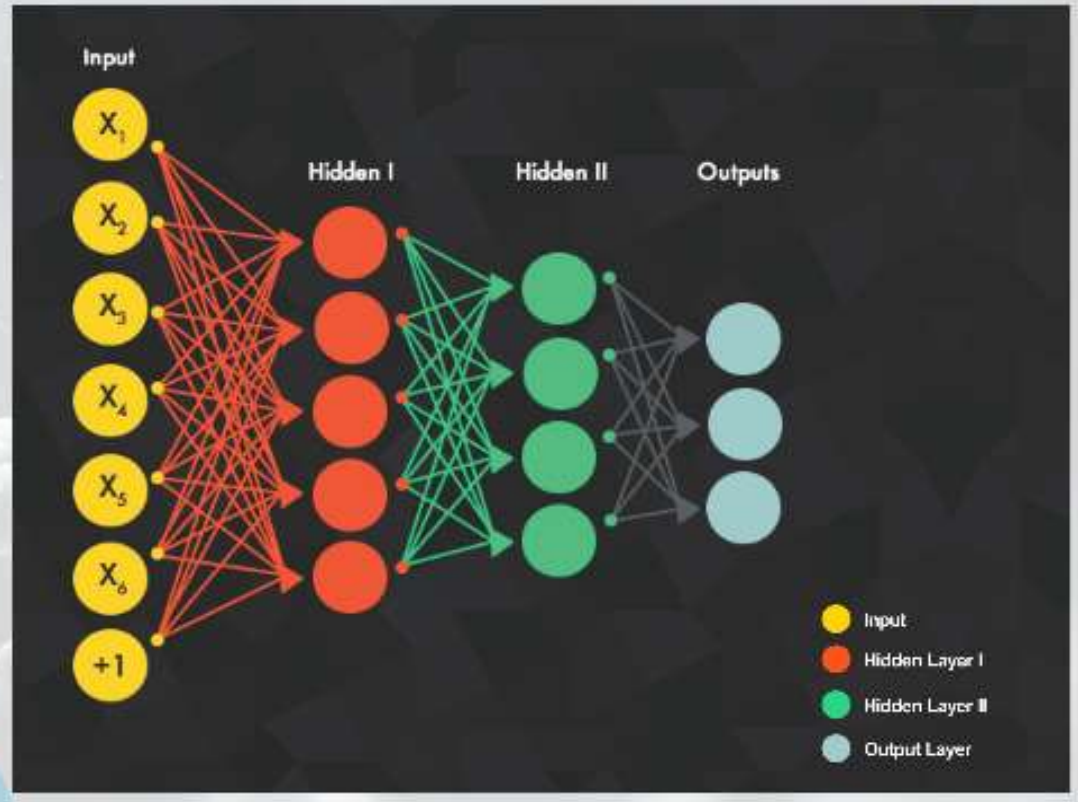
3. To bring to the fore the various research issues and trend in deep learning.

What is
Deep Learning

What Is Deep Learning?

- A class of machine learning techniques that exploit many layer of non linear information processing for supervised or unsupervised feature extraction and trasformation and for pattern analysis and classification (Deng & Yu, 2014)

- It typically uses deep artificial neural networks.



What is deep Learning?

- A sub-field of machine learning that is based on learning several levels of representations corresponding to a hierarchy of features or factors or concepts, where higher-level concepts are defined from lower-level ones, and the same lower level concepts can help to define many higher-level concepts.

(LeCun, Bengio, & Hinton, 2015).

▪Deep Learning = Machine Learning + Representation Learning

- *Representation learning is a set of methods that allows a machine to be fed with raw data and to automatically discover the representations needed for detection or classification.*(Deng & Yu, 2014).

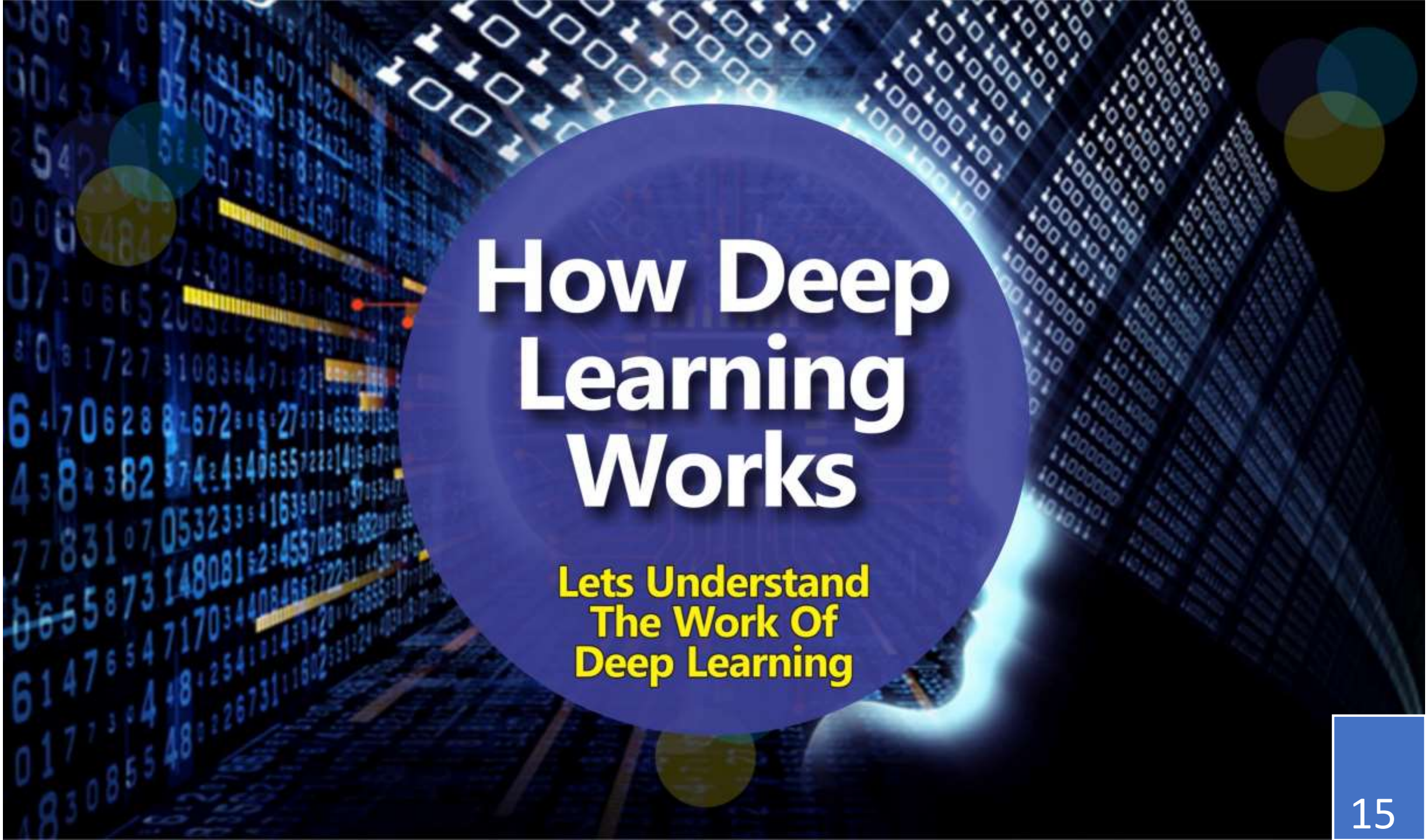
Why Is Deep Learning Happening Now?

- Actually it is not - first papers published in '970s
- Third resurgence of neural networks
  - Research breakthroughs

	Problem	Solution
Vanishing Gradient	Sigmoid type activation functions easily saturate. Gradients are small. in deeper layers updates become almost zero.	Earlier: layer-by-layer pretraining Recently: non-saturating Activation functions
Gradient Descent	First order methods (e.g. SGD) are easily stuck. Second order methods are feasible on larger data.	Adaptive training: adagrad, adam, adadelta, RMSProp Nestrov momentum
Regularisation	Networks easily overfit (even with l2 regularization	Dropout

  - Increase in computational power
  - GP GPUs

Adapted from: www.edureka.co


How Deep Learning Works
Lets Understand The Work Of Deep Learning

How Deep Learning Works?

- Deep Learning Is Implemented Through Neural Networks
- Motivation Behind Neural Networks Is The Biological Neuron.

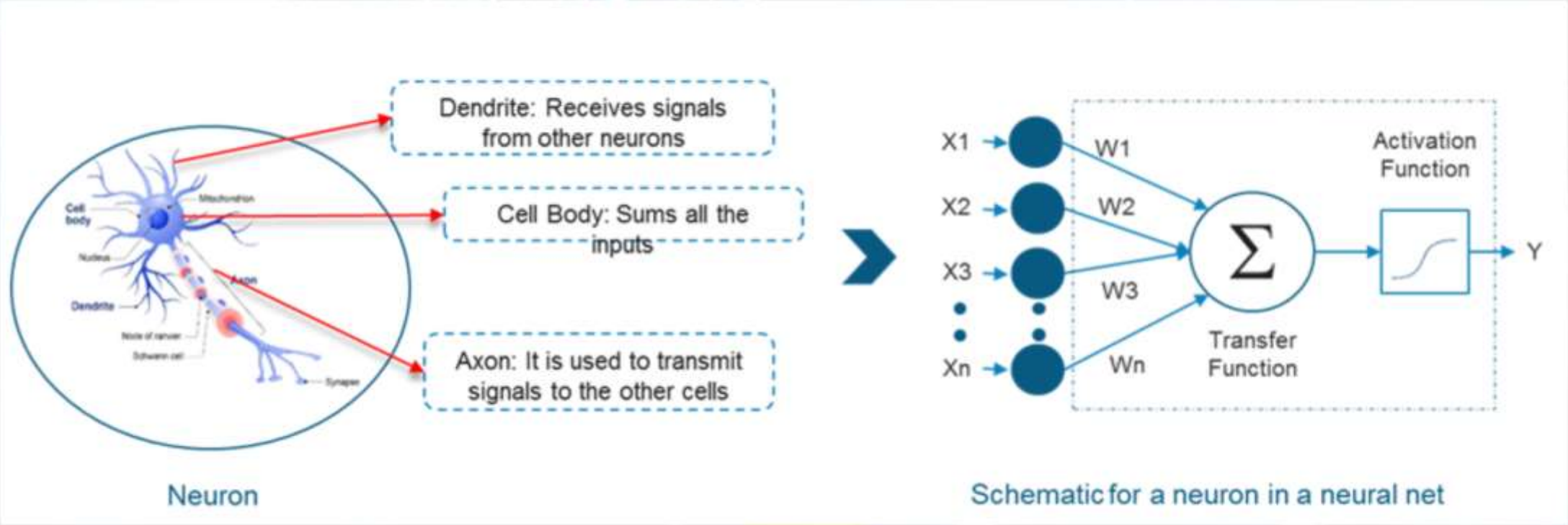


Figure 7: Biological and Artificial Neuron (Adapted from: www.edureka.co)

Unique Characteristics Of Deep Learning

- Automatic feature extraction
- Makes use of High-dimensional data for training
- Defines higher level features from lower level features obtained from the previous layer
- It is a representation learning method
- Can solve real-world or unconstrained problems such as NLP, image recognition etc

Some Deep Learning Software

The Following Are The List Of Some Deep Learning Software's But Not Limited To:

- Tensorflow
- Theano
- Torch
- Keras
- Blocks
- Caffe
- Caffe2
- MxNet

- MatConvNet
- Microsoft Cognitive Toolkit
- Deep learning 4j
- Dlib

...but not limited to these.

Categories of Deep Architecture with Algorithms

1. Supervised or Discriminative Deep Learning
2. Unsupervised or Generative Deep Learning
3. Hybrid Deep Architecture

1 Supervised or Discriminative Deep Learning

- **Convolutional Neural Networks (CNNs)**

A subtype of the discriminative architecture is the CNN.

- The Convolutional Neural Networks (CNN) is one of the most notable deep learning approaches where multiple layers are trained in a robust manner (LeCun, Bottou, Bengio, & Haffner, 1998).

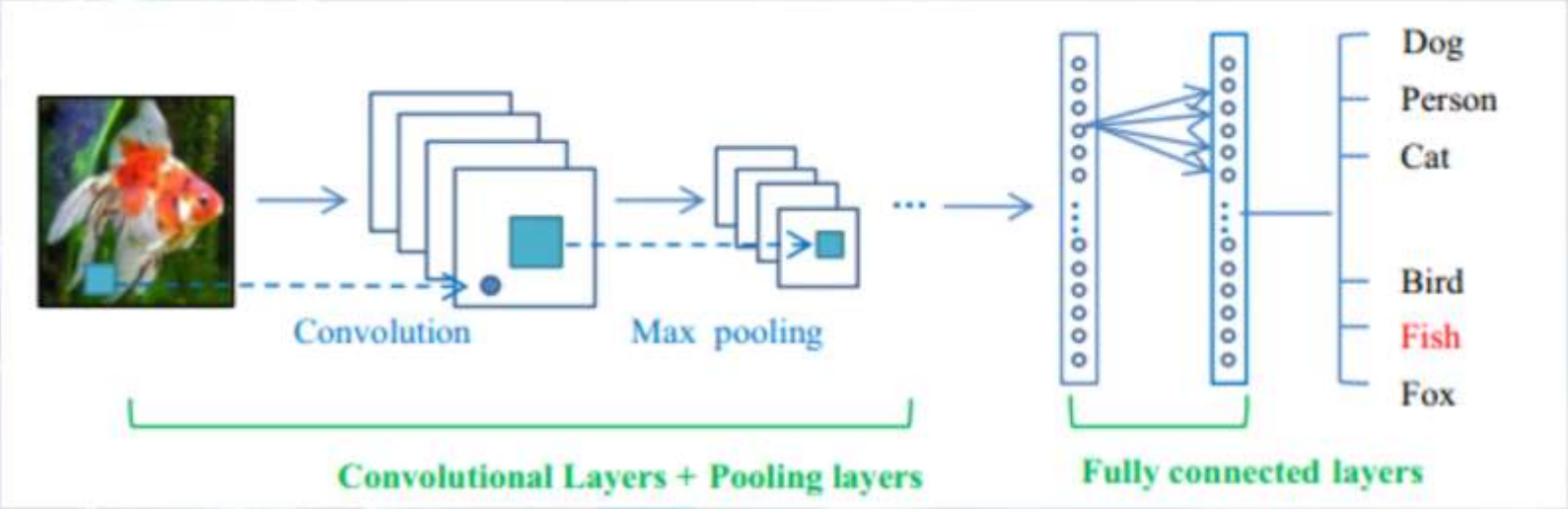


Figure 8: Pipeline of the general CNN Architecture (Source: Guo et al., 2016)

CNN in Action: Facial Recognition

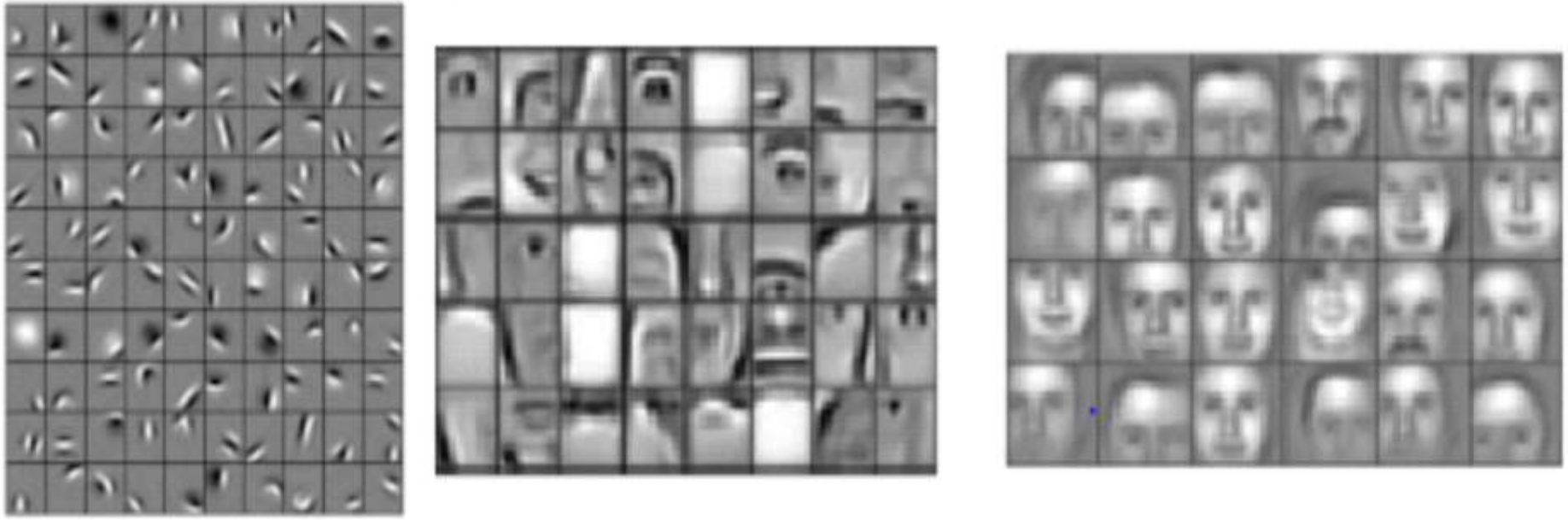


Figure 9: Learned hierarchical features from a deep learning algorithm, CNN
(Source: lee et al., 2011)

2 Unsupervised or Generative Deep Learning

Restricted Boltzmann Machines (RBM)

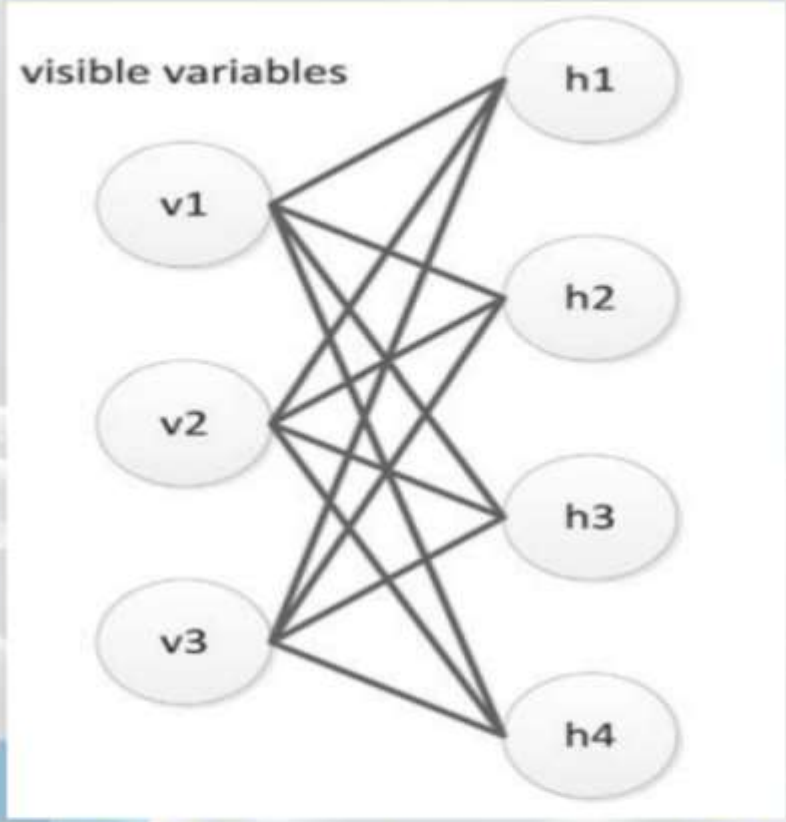


Figure 10 : Schematic structure of RBM (Source: Guo et al., 2016)

Variants of Restricted Boltzmann Machines (RBM)

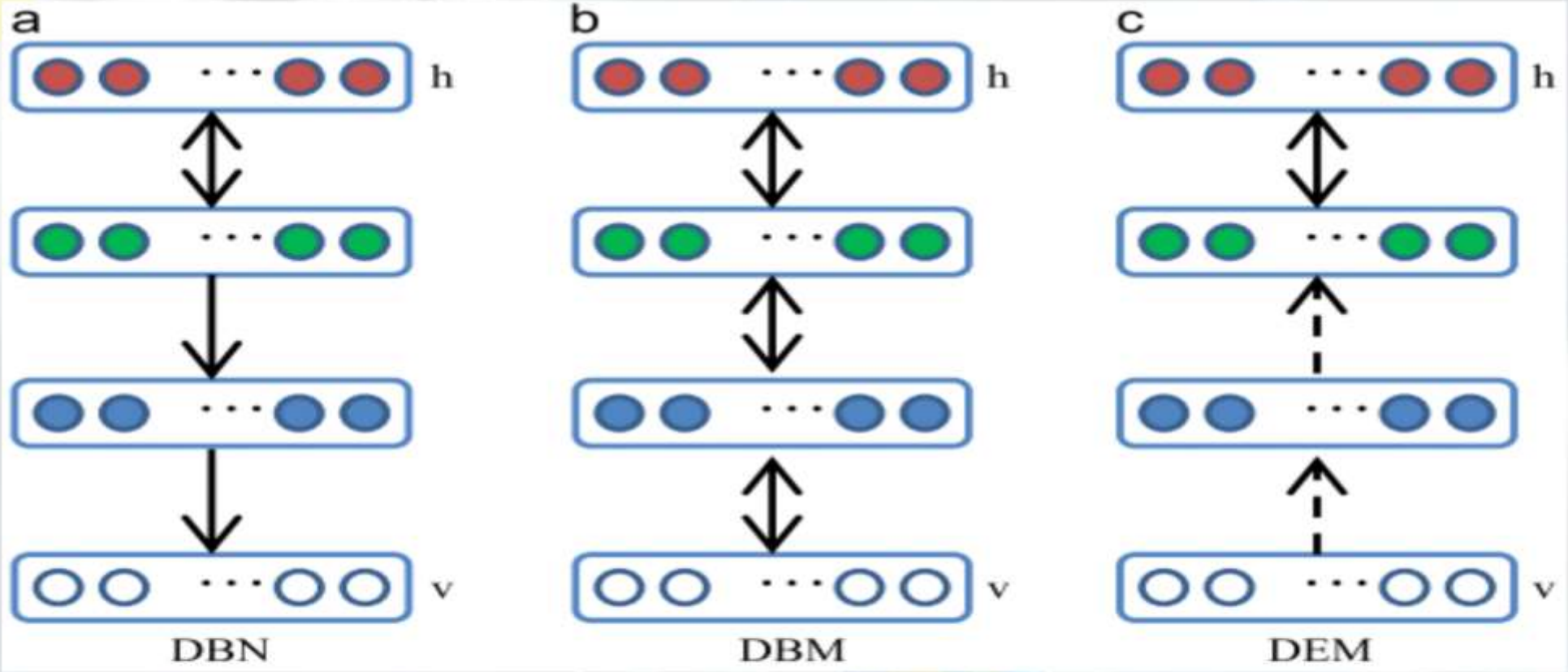


Figure 11: DBN, DBN and DEM (Source: Guo et al., 2016).

Auto-Encoder (AE)

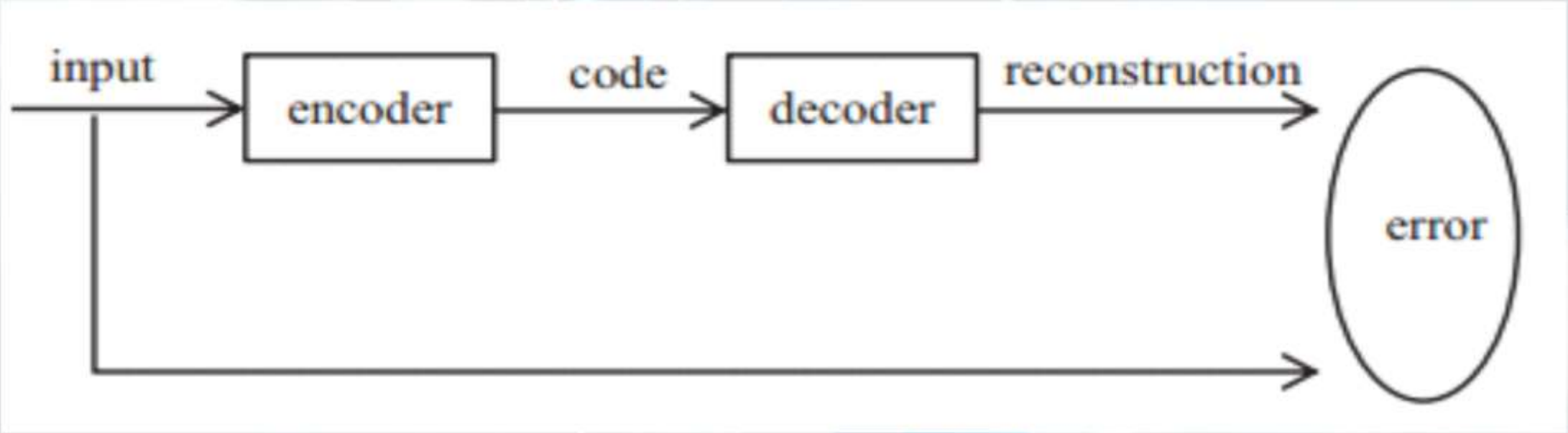
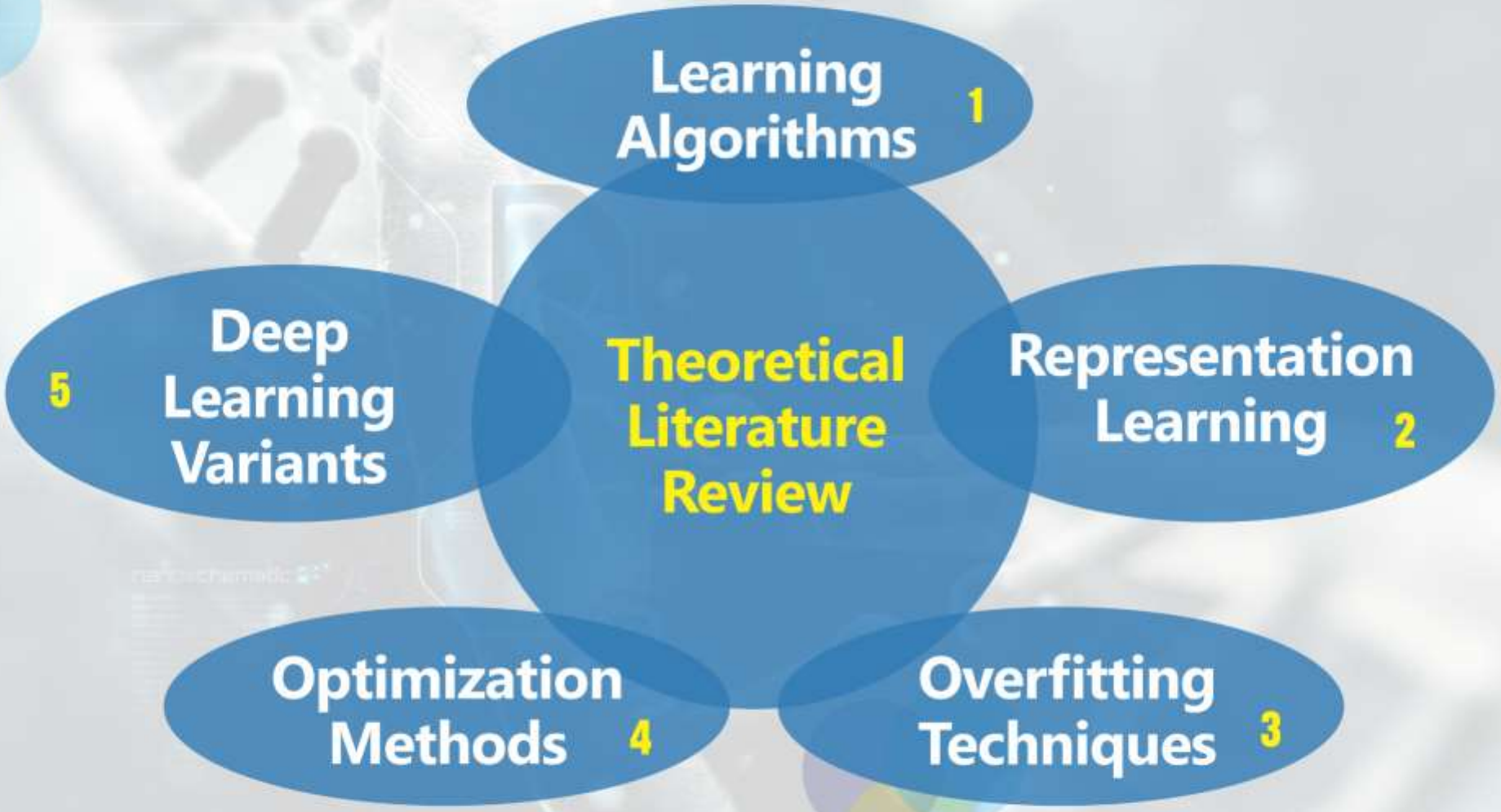


Figure 12: The pipeline of an autoencoder (Source: Guo et al., 2016).


Theoretical Literature Review
Learning Algorithms 1
Representation Learning 2
Overfitting Techniques 3
Optimization Methods 4
Deep Learning Variants 5

(1) Learning Algorithms

Table 1: A categorization of the basic deep NN learning algorithms and related approaches.(Source: Guo **et al., 2016).**

CNN	RBM	AUTOENCODER	SPARSE CODING
AlexNet (Krizhevsky et al, 2012)	Deep Belief Net (Hinton, et al, 2006)	Sparse Autoencoder (Poultney et al 2006)	Sparse Coding (Yang et al, 2009)
Clarifai (Zeiler, et al 2014)	Deep Boltzmann Machine (Salakhutdinov et al., 2009)	Denoising Autoencoder (Vincent, et al. 2008)	Laplacian Sparse coding (Gao et al, 2010)
SPP (He et al, 2014)	Deep Energy Models (Ngiam et al., 2011)	Contractive Autoencoder (Rifai, et al.2011)	Local Co-ordinate coding (Yu et al, 2009)
VGG (Simonyan et al., 2014)			Super-Vector coding (Zhou et al, 2010)
GoogLeNet (Szegedy et al., 2015)			

2. Representation Learning

S/N	Research Focus	Contribution
1	To discover a fast and more efficient way of initializing weights for effective learning of low-dimensional codes from high-dimensional data in multi-layer neural networks (**Hinton et al., 2006; Salakhutdinov et al., 2009; Vincent et al., 2010; Cho et al., 2011).**	Implementation of a novel learning algorithm for initializing weights that allows deep AE networks and deep boltmzann machines to learn useful higher representations
2	To explore the possibility of allowing hashing function learning(learning of efficient binary codes that preserve neighborhood structure in the original data space) and feature learning occur simultaneously (**Salakhutdinov et al., 2009; Erin et al., 2015; Zhong et al., 2016).**	Introduction of a state–of-the-art deep hashing, supervised deep hashing and semantic hashing methods for large scale visual search, image retrieval and text mining
3	To bridge the gap between the success of CNNSs for supervised learning and unsupervised learning (**Springenberg et al., 2014; Radford et al., 2015**).	Introduction of a class of CNNs called deep convolutional generative adversarial networks (DCGANs) for unsupervised learning.

3. Overfitting Techniques

S/N	Research Focus	Contribution
1	To mitigate the problem of overfitting in large neural networks with sparse datasets (**Zeiler et al., 2013; Srivastava et al., 2014; Pasupa et al., 2016;).**	Implementation of several regularization techniques such as "dropout", stochastic pooling, weight decay, flipped image augmentation amongst others for ensuring stability in DNN
2	To investigate how to automatically rank source CNNs for transfer learning and use transfer learning to improve a Sum-Product Network for probabilistic inference when using sparse datasets (**Afridi et al., 2017; Zhao et al., 2017).**	Design of a reliable theoretical framework that perform zeroshot ranking of CNNs for transfer learning for a given target task in Sum-Product networks

4. Optimization Methods

S/N	Research Focus	Contribution
1	To develop a computationally efficient algorithm for gradient based optimization in deep neural networks (**Hinton et al., 2006; Duchi et al., 2011; Ngiam et al., 2011; Tieleman et al., 2012; Sutskever et al., 2013; Kingma et al., 2014; Patel, 2016).**	Introduction of several first-order and second-order stochastic gradient based optimization methods for minimizing large objective functions in deep networks such as Complementary priors, Adam, Adagrad, RMSprop, Momentum, L-BFGS and Kalman- based SGD
2	To develop an accelerator that can deliver state-of-the-art accuracy with minimum energy consumption when using large CNNs(**Chen et al., 2017).**	Implementation of an Energy-Efficient Reconfigurable Accelerator for Deep CNN using efficient dataflow to minimize energy through zeros skipping/gating.

5. Deep Learning Variants

S/N	Research Focus	Contribution
1	To demonstrate the advantage of combining deep neural networks with support vector machines (**Zhong et al., 2000; Nagi et al., 2012; Tang et al., 2013; Li et al., 2017).**	Introduction of a novel classifier architecture that combines **two heterogeneous supervised classification techniques**, CNN and SVM for feature extraction and for classification
2	To exploit the power of deep neural networks in optimizing the performance of nearest neighbor classifiers in kNN Classification tasks (**Min et al., 2009; Ren et al., 2014).**	They presented a framework for learning convolutional nonlinear features for K nearest neighbor (kNN) classification.

Application Literature Review

1
Methodology
Application of DL

2
Domain
Application of DL

Deep learning has been applied in so many ways to solve real life problems among which are:

(1) Methodology Application of DL

Author and Title	Objective	Methodology	Contribution
Araque et al.(2017). Enhancing deep learning sentiment analysis with ensemble techniques in social applications.	To improve the performance of sentiment analysis in social applications by integrating deep learning techniques with traditional feature based approaches based on hand-crafted or manually extracted features	The utilization of a word embedding's model and a linear machine learning algorithm to develop a deep learning based sentiment classifier(baseline), the use of two ensemble techniques namely ensemble of classifiers (CEM) and ensemble of features (M_{SG} and M_{GA})	Development of ensemble models for sentiment analysis which surpass that of the original baseline classifier

Methodology Application of DL (Contd)

Author and Title	Objective	Methodology	Contribution
Betru et al. (2017). Deep Learning Methods on Recommender System. A Survey of State-of-the-art:	To distinguish between the various traditional recommendation techniques and introducing deep learning collaborative and content based approaches	As pointed out by the authors, the methodology adopted in (Wang, Wang, & Yeung, 2015) integrated a Bayesian Stack De-noising Auto Encoder (SDAE) and Collaborative Topic Regression to perform collaborative deep learning.	The implementation of a novel collaborative deep learning approach, the first of its kind to learn from review texts and ratings.

Methodology Application of DL (Contd)

Author and Title	Objective	Methodology	Contribution
Luo et al.(2016). A deep learning approach for credit scoring using credit default swaps.	To implement a novel method which leverages a DBN model for carrying out credit scoring in credit default swaps (CDS) markets	The methodology adopted by the researchers in their experiments was to compare the results of MLR, MLP, and SVM with the Deep Belief Networks (DBN) with the Restricted Boltzmann Machine by applying 10-fold cross-validation on a dataset	The contribution made by the researchers to this literature is investigating the performance of DBN in corporate credit scoring. The results demonstrate that the deep learning algorithm significantly outperforms the baselines.

Methodology Application of DL (Contd)

Author and Title	Objective	Methodology	Contribution
Grinblat et al.(2016). Deep learning for plant identification using vein morphological patterns.	The authors aimed to eliminate the use of handcrafted features extractors by proposing the use of deep convolutional network for the problem of plant identification from leaf vein patterns.	The methodology adopted to classify three plant species: white bean, red bean and soybean was the use of dataset containing leaf images, a CNN of 6 layers trained with the SGD method, a training set using 20 samples as mini batches with a 50% dropout for regularization.	The relevance of deep learning to agriculture using CNN as a model for plant identification based on vein morphological pattern.

Methodology Application of DL (Contd)

Author and Title	Objective	Methodology	Contribution
Evermann et al.(2017). Predicting process behaviour using deep learning.	To come up with a novel method of carrying out process prediction without the use of explicit models using deep learning.	The approach used to implement this novel idea included the use of a framework called Tensorflow as it provides (RNN) functionality embedded with LSTM cells which can be run on high performance parallel, cluster and GPU platforms.	Improvement in state-of-the-art in process prediction, the needless use of explicit model and the inherent advantages of using an artificial intelligence approach.

Methodology Application of DL (Contd)

Author and Title	Objective	Methodology	Contribution
Kang et al. (2016). A deep-learning-based emergency alert system.	proposed a deep learning emergency alert system to overcome the limitations of the traditional emergency alert systems	A heuristic based machine learning technology was used to generate descriptors starts for labels in the problem domain, an API analyzer that utilized convolutional neural network for object detection and parsing to generate compositional models was also used.	Contribution of this research shows that the EAS can be adapted to other monitoring devices asides from CCTV

(2) Domain Applications Of Deep Learning

Domain	Deep learning is Applied to perform	Topic &Reference
Recommender System	Sentiment Analysis/Opinion mining)	Collaborative Deep Learning for Recommender Systems(Wang, Wang, & Yeung, 2015)
Social Applications	(Sentiment Analysis/Opinion mining/Facial Recognition)	Enhancing deep learning sentiment analysis with ensemble techniques in social applications (Araque et al., 2017).
Medicine	(Medical Diagnosis)	A survey on deep learning in medical image analysis(Litjens et al., 2017)
Finance	(Credit Scoring, stock market prediction)	A deep learning approach for credit scoring using credit default swaps (Luo et al., 2016).

Domain Applications 0f Deep Learning (Contd)

Domain	Deep learning is Applied to perform	Topic &Reference
Transportation	Traffic flow prediction	Deep learning for short-term traffic flow prediction(Polson et al.,2017).
Business	Process prediction	Predicting Process Behaviour Using Deep Learning (Evermann et al., 2017).
Emergency	Emergency Alert	A Deep-Learning-Based Emergency Alert System (Kang et al., 2016)
Agriculture	(Plant Identification)	Deep Learning for Plant Identification Using Vein Morphological Patterns (Grinblat et al.,2016).

Applications of Deep Learning

Some Amazing & Recent Application Of Deep Learning Are:

- Automatic Machine Translation
- Object Classification In Photographs
- Character Text Generation
- Colorization Of Black And White Images
- Automatic Game Playing

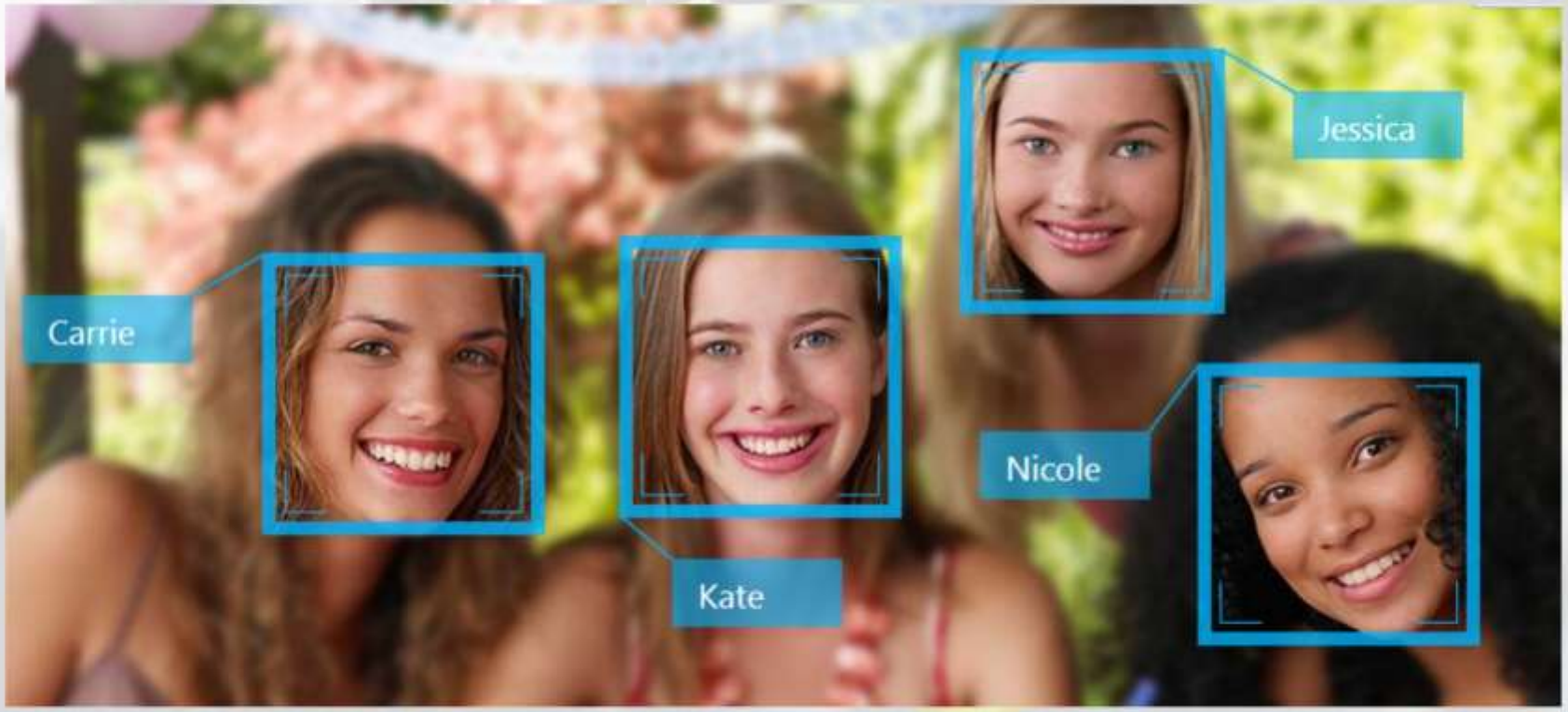
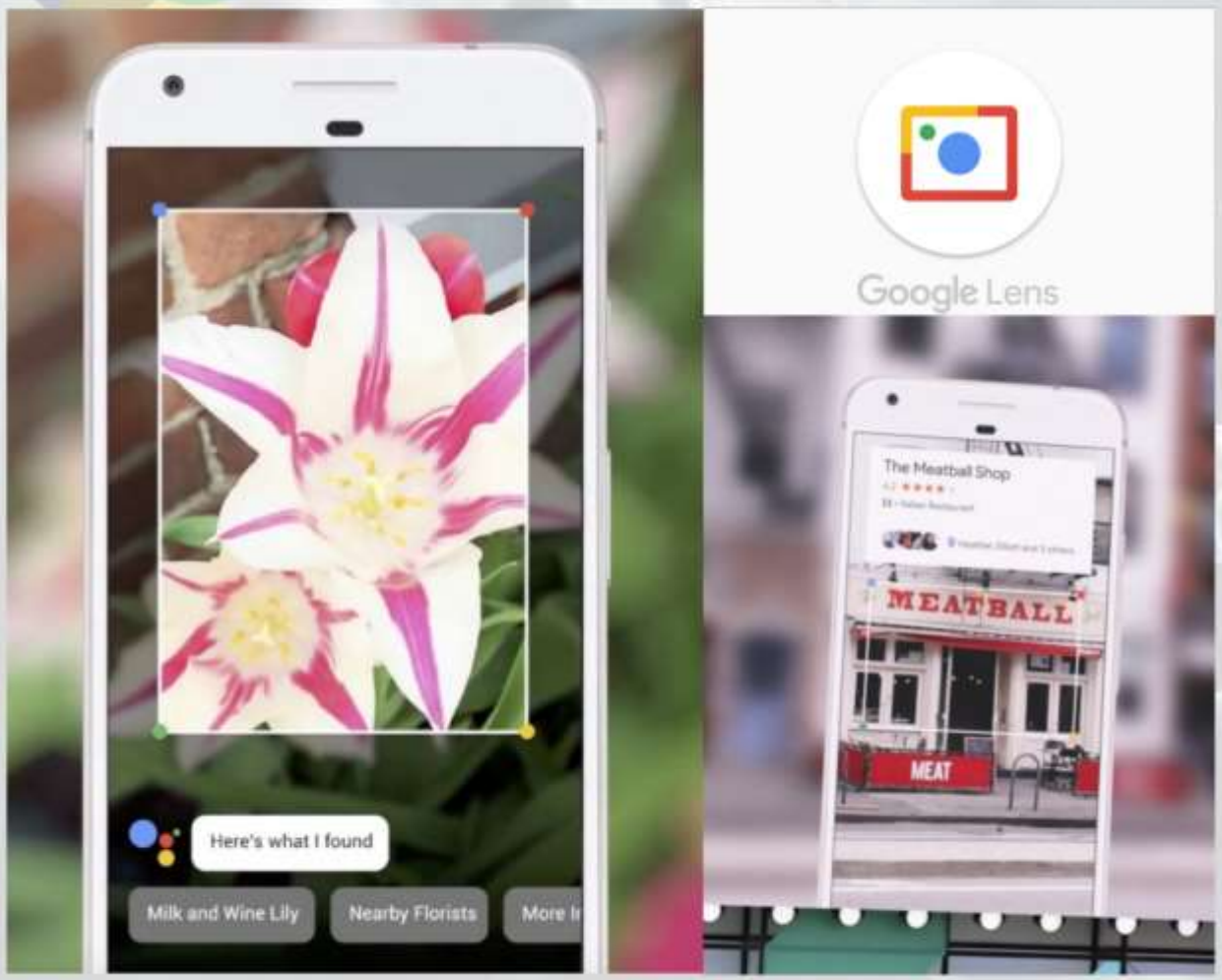


Figure 13: Face Recognition (Adapted from www.edureka.co)

Applications of Deep Learning (Contd)



- Google lens is a set of vision based computing capabilities that allows your smartphone to understand what's going on in a photo, video or a live feed.
- For instance, point your phone at a flower and Google Lens will tell you on the screen which type of flower it is.
- You can aim the camera at a restaurant sign to see reviews and other information.

Figure 14: Google Lens (Adapted from www.edureka.co)

Applications of Deep Learning (Contd)

Translation

- This is a situation where you are given words in some language and you have to translate the words to desired language say english
- This kind of translation is a classical example of image recognition

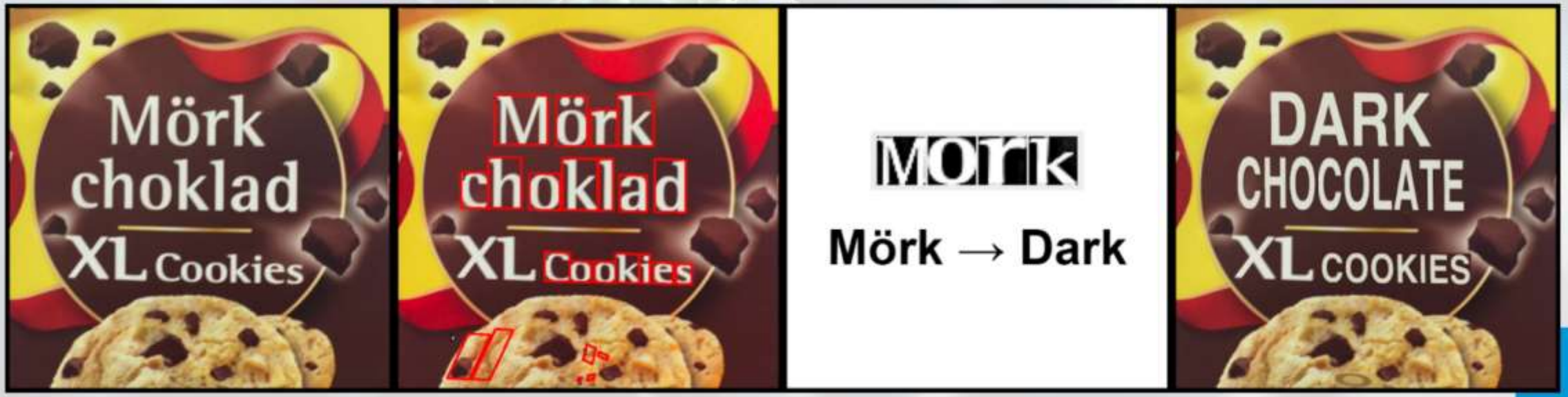


Figure 15: Machine Translation (Adapted from www.edureka.co)

Instant Visual Translation

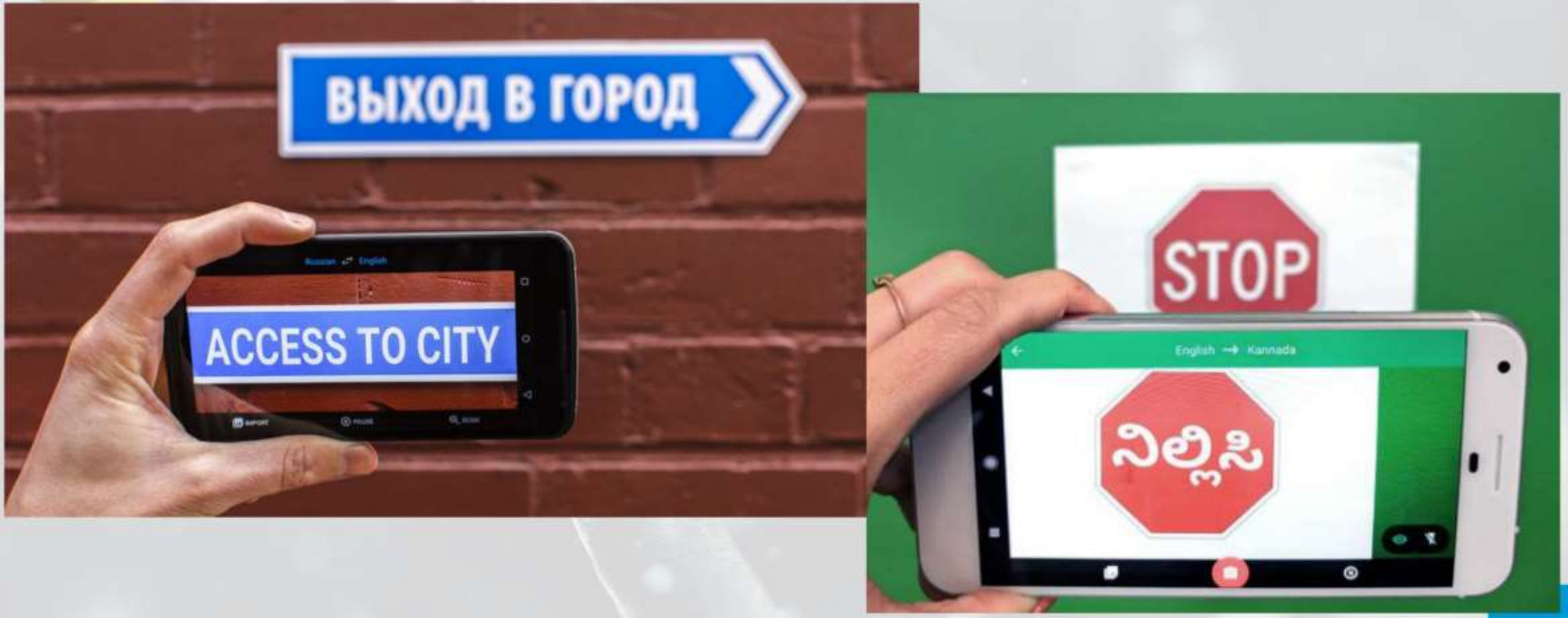


Figure 16: Instant Visual Translation (Adapted from www.edureka.co)

Self Driving Cars

Figure 17: Self Driving Cars (Adapted from www.edureka.co)

Automatic Machine Translation

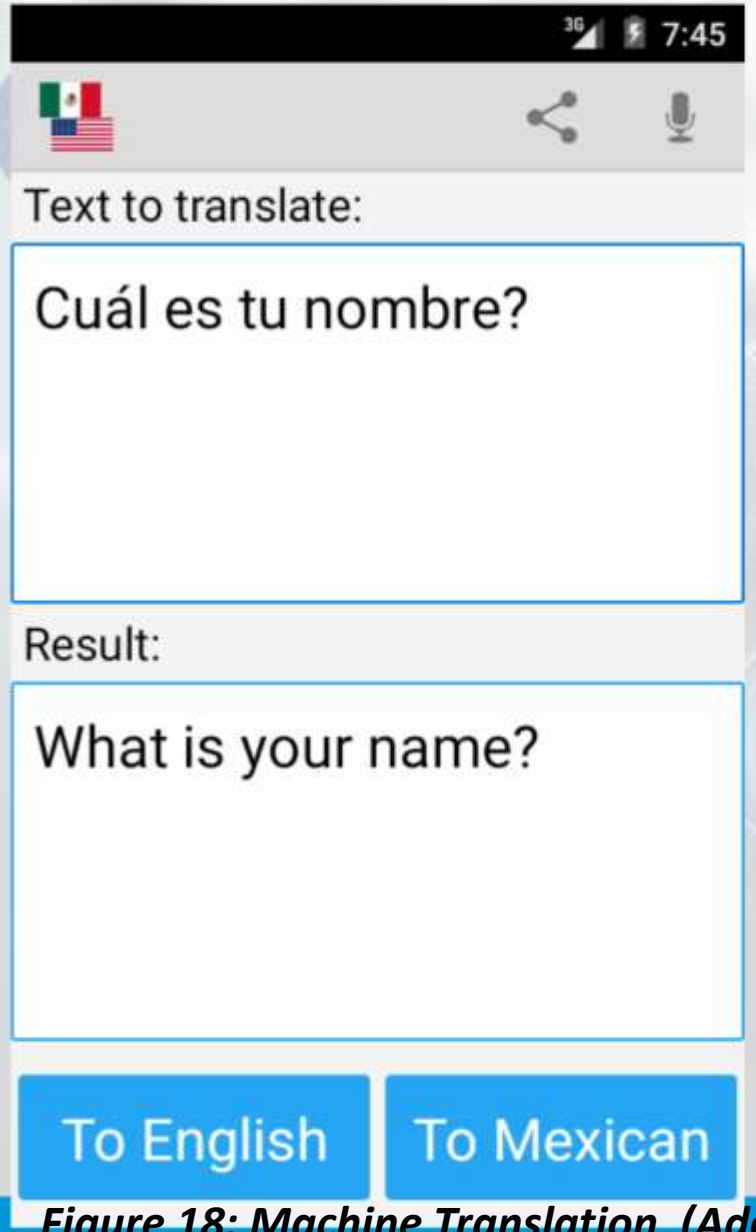


Figure 18: Machine Translation (Adapted from www.edureka.co)

Trends in Deep Learning Research

1. Design of more powerful deep models to learn from fewer training data. (Guo et al, 2016; pasupa et al., 2016 ;Li, et al 2017)
2. Use of better optimization algorithms to adjust network parameters i.e. regularization techniques (zeng et al, 2016; Li, et al 2017)
3. Implementation of deep learning algorithms on mobile devices (Li, et al 2017)
4. Stability analysis of deep neural network (Li, et al 2017)

Trends in Deep Learning Research (Contd)

5. Combining probabilistic , auto-encoder and manifold learning models.(bengio et al., 2013)
6. Applications of deep neural networks in nonlinear networked control systems (NCSs) (Li, et al 2017)
7. Applications of unsupervised, semi-supervised and reinforcement-learning approaches to DNNs for complex systems (Li, et al 2017)
8. Learning deep networks for other machine learning techniques e.g. deep kNN (Zoran et al, 2017), deep SVM (Li, et al 2017).

Research Issues/challenges in Deep Learning

1. High Computational cost/burden in training phase (pasupa et al., 2016)
2. Over-fitting problem when the data-set is small. (pasupa et al., 2016, Guo et al, 2016)
3. Optimization issues due to local minima or use of first order methods. (pasupa et al., 2016)
4. Little or no clear understanding of the underlying theoretical foundation of which deep learning architecture should perform well or outperform other approaches. (Guo et al, 2016)
5. Time complexity (Guo et al, 2016)

Deep Learning – Use Case

Let’s look at a use case where we can use DL for image recognition

Practical Application of deep learning in Facial Recognition

Problem Scenario

Suppose we want to create a system that can recognize faces of different people in an image. How do we solve this as a typical machine learning problem and/or using a deep learning approach?

Classical Machine Learning Approach

We will define facial features such as eyes, nose, ears etc. and then, the system will identify which features are more important for which person on its own or by itself.

Deep Learning Approach

Now, deep learning takes this one step ahead.

Deep learning automatically finds out the features which are important for classification because of deep neural networks, whereas in case of Machine Learning we had to manually define these features.

Practical Application of deep learning - Facial Recognition (Contd)

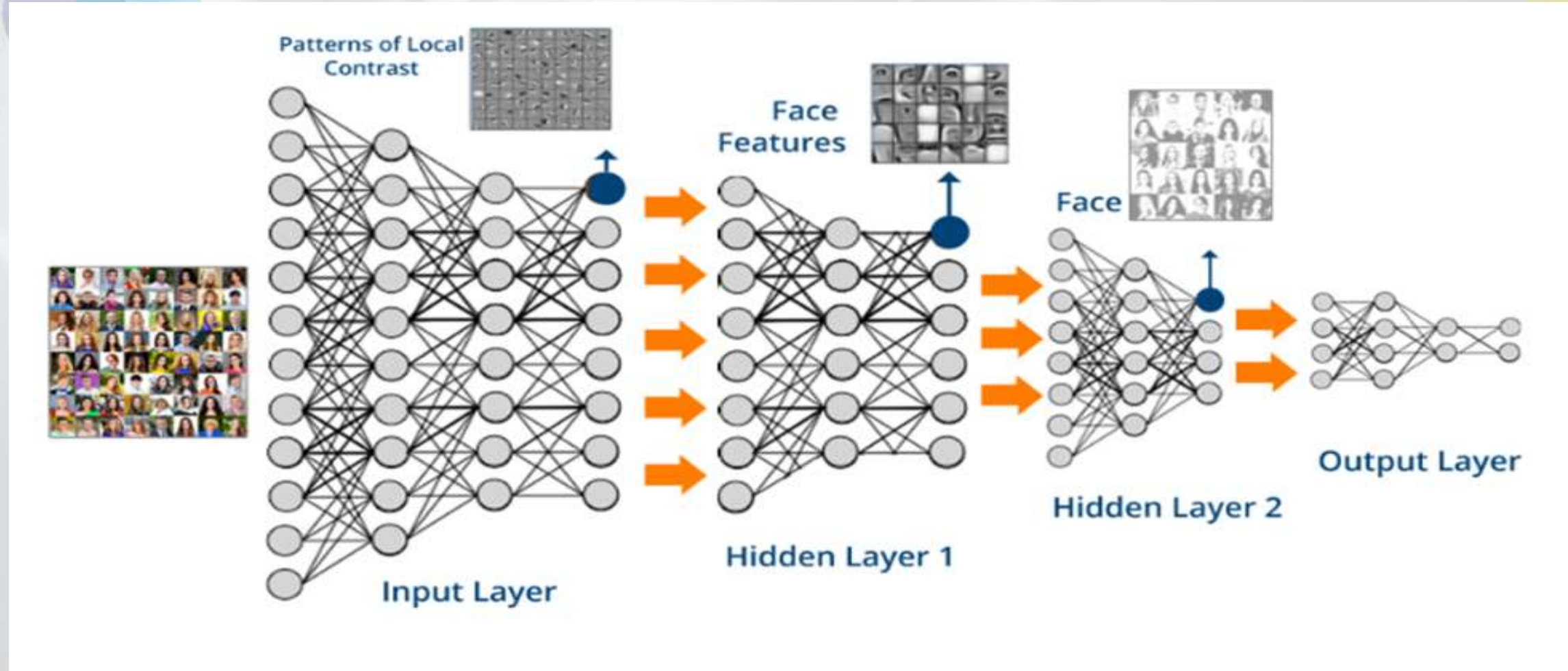


Figure 19: Face Recognition Using deep networks (Source: www.edureka.co)

Deep Face Recognition

Face recognition applications have two parts or phases viz:

(1) Phase-I: Enrollment phase – Model / system is trained using millions of prototype face images and a trained model is generated. Generated face features are stored in database and

(2) Phase-II: Recognition phase – Query face image is given as input to the model generated in phase-I to recognise it correctly.

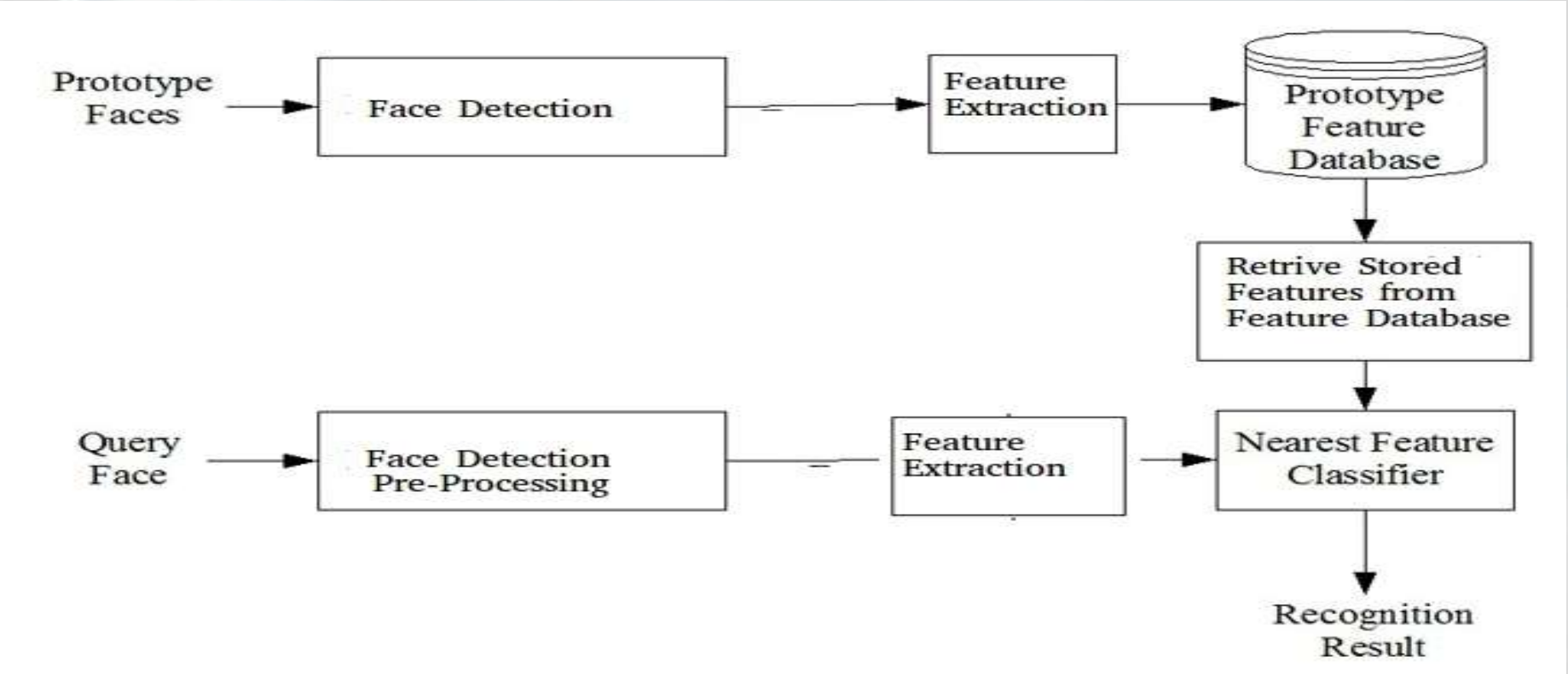


Figure 20: Face Recognition Architecture (Source: aiehive.com)

Deep Face Recognition (Contd)

Steps within Enrollment Phase Includes

1. Face Detection
2. Feature extraction
3. Store Model and extracted feature in Database

Steps within Recognition Phase / Query Phase Includes

1. Face Detection
2. Preprocessing
3. Feature Extraction
4. Recognition

Step 1: Face Detection - Enrollment Phase

Face Detection: Face needs to be located and region of interest is computed.

- Histogram of Oriented Gradients (HOG) is a faster and easier algorithm for face detection.
- Detected faces are given to next step of feature extraction.

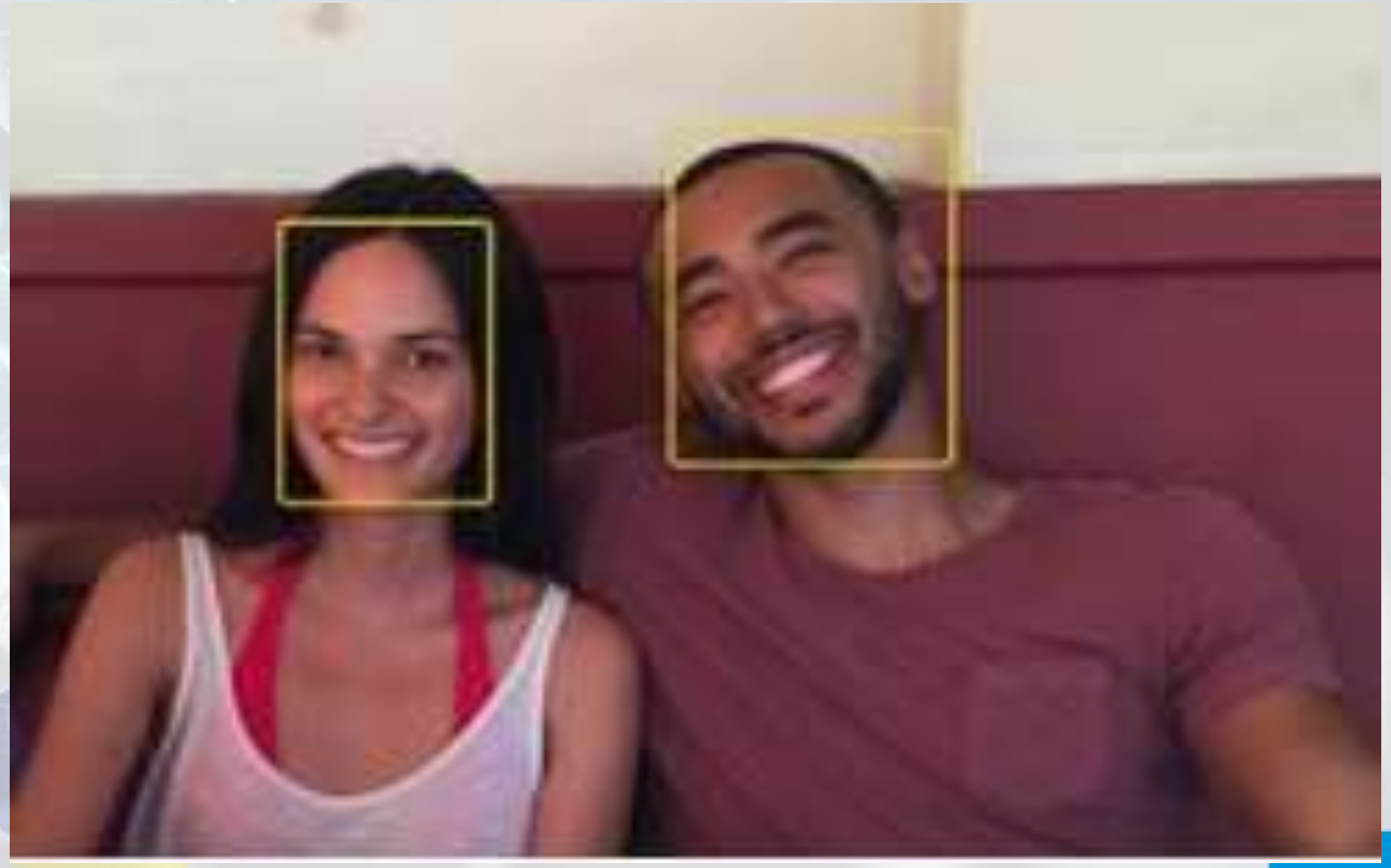
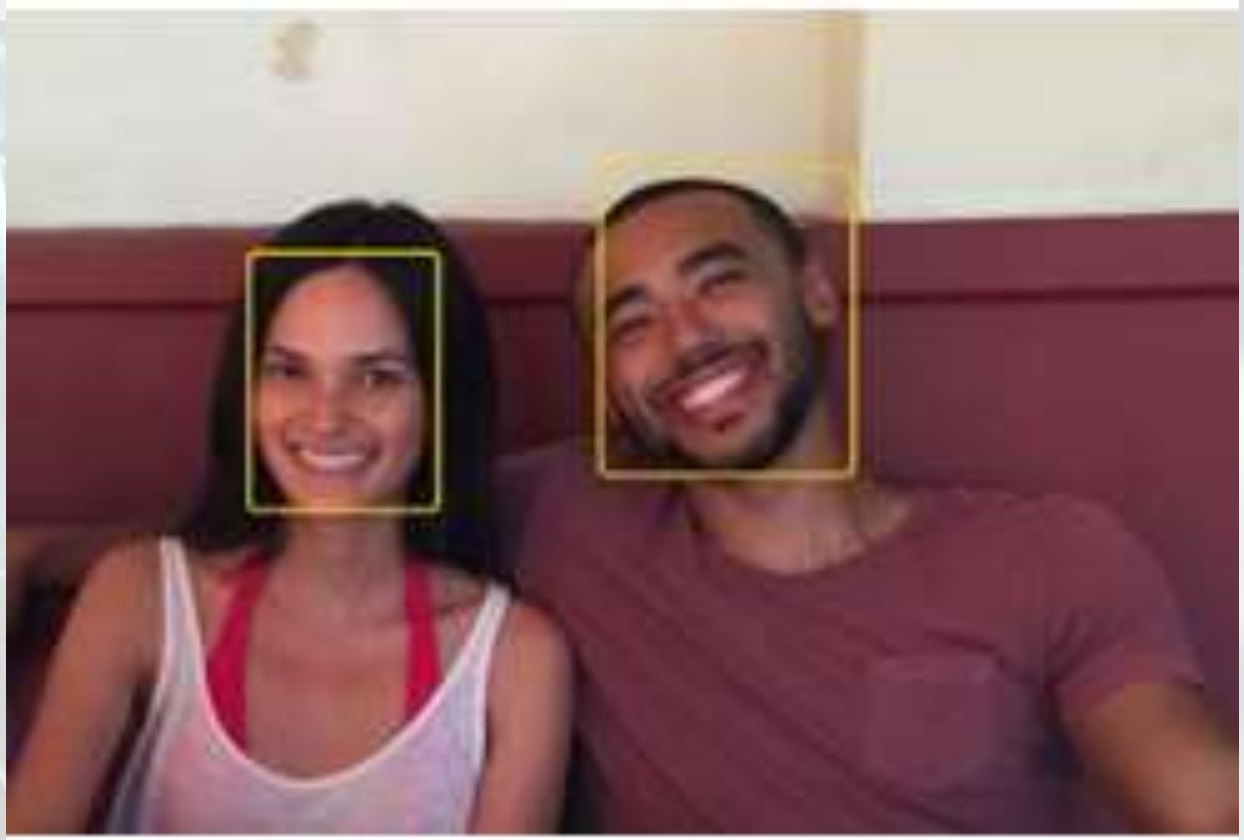


Figure 21: Multiple Face Detection (Source: aiehive.com)

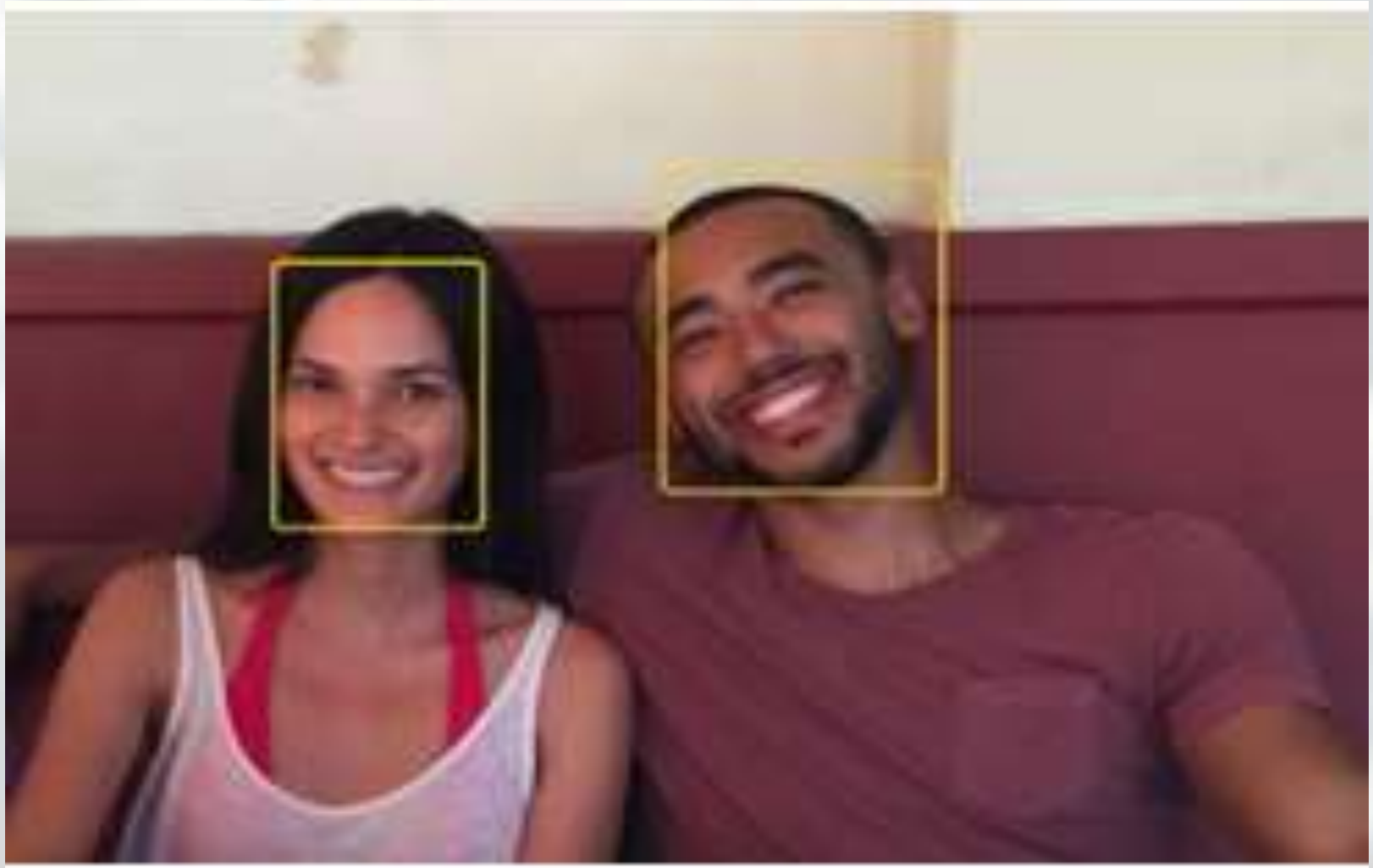
Step 2: Feature Extraction- Enrollment Phase

What is the best feature measure that represents human face in a best way?

Deep learning can determine which parts of a face are important to measure. Deep Convolution Neural Network (DCNN) can be trained to learn important features. (Simonyan et al., 2014)



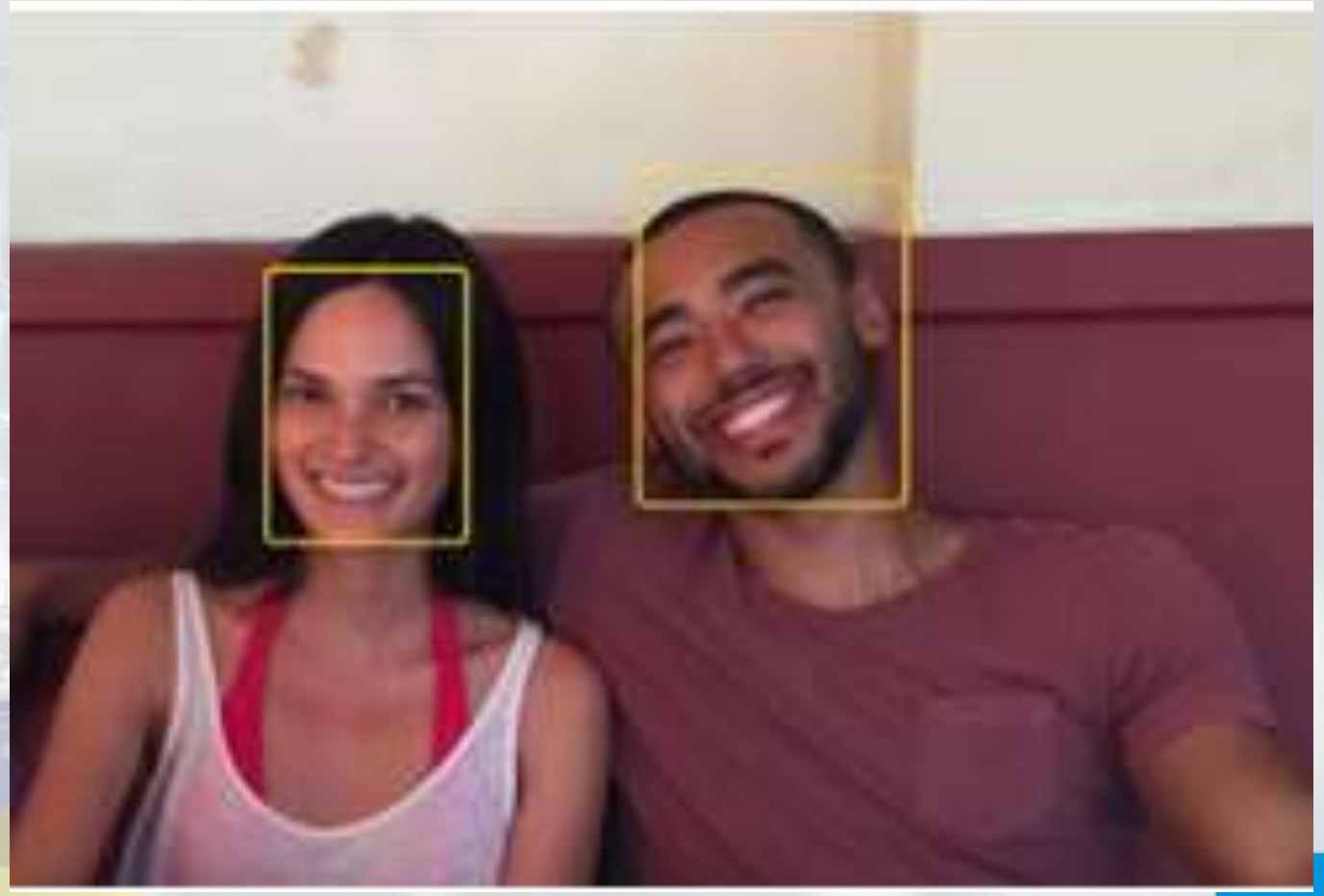
Step-3. Store DCNN model and Feature in Database- Enrollment Phase



Step 1: Face Detection- **Recognition Phase**

Face Detection: Face needs to be located and region of interest is computed.

- Histogram of Oriented Gradients (HOG) is faster and easier algorithm for face detection.
- Detected faces are given to next step of preprocessing.



Step-2. Pre-processing- Recognition Phase

- Pre-process to overcome issues like noise, illumination using any suitable filters [Kalman Filter, Adaptive Retinex (AR), Multi-Scale Self Quotient (SQI), Gabor Filter, etc.]
- Pose/rotation can be accounted by using 3D transformation or affine transformation or face landmark estimation
- Determine 68 landmark points on every face — the top of the chin, the outside edge of each eye, the inner edge of each eyebrow, etc.

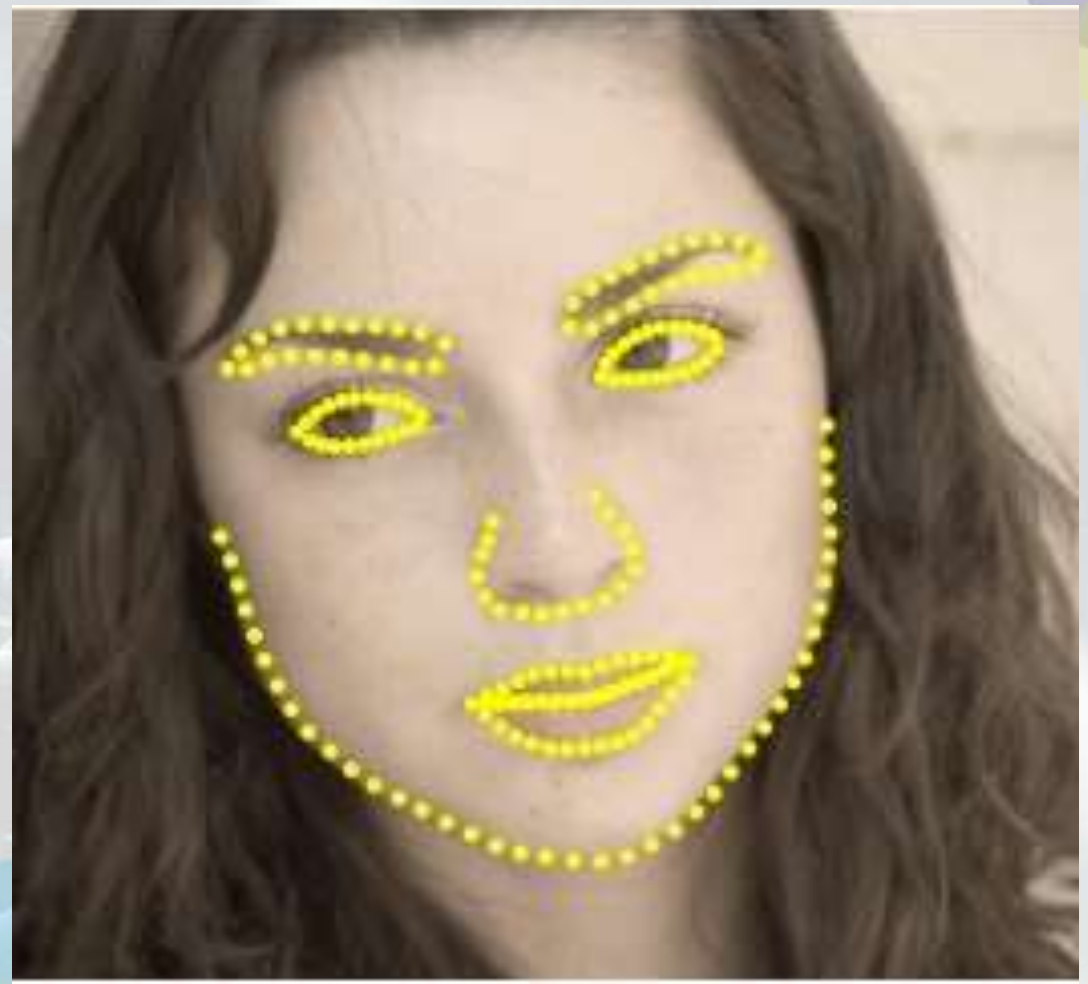


Figure 22: **Landmark point estimation *(Source: aiehive.com)***

Step 3: Feature Extraction-Recognition Phase

- In this third step of Deep Face Recognition, we have to use trained DCNN model, which was generated during feature extraction step of enrollment phase
- A query image is given as input.
- The DCNN generates 128 feature values.
- This feature vector is then compared with feature vector stored in database

Step 4: Recognition-Recognition Phase

- This can be done by using any basic machine learning classification algorithm SVM classifier, Bayesian classifier, Euclidean Distance classifier, for matching database feature vector with query feature vector.

- Gives ID of best matching face image from database as a recognition output.

Conclusion

- Deep learning is a representation learning method and the new state-of-the-art technique for performing automatic feature extraction in large unlabeled data
- Various categories of deep learning architectures and basic algorithms together with their related approaches have been discussed
- Several theoretical concepts and practical application areas have been presented
- It is a promising research area for tackling feature extraction for complex real-world problems without having to undergo the process of manual feature engineering.
- With the rapid development of hardware resources and computation technologies, it is certain that deep neural networks will receive wider attention and find broader applications in the future.

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